

Multi-label Image Classification on PASCAL VOC 2007

Benchmarking CNN architectures — from scratch and pretrained — on 20-class multi-label detection with mean Average Precision.

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Overview

PASCAL VOC 2007 contains approximately 10,000 images across 20 object categories; each image may contain multiple classes simultaneously, making this a multi-label classification problem rather than a single-label one. Performance is measured by mean Average Precision (mAP) — the average of per-class AP scores across all 20 categories. This project systematically compares four CNN models: AlexNet trained from scratch, ImageNet-pretrained AlexNet fine-tuned on VOC, a compact custom SimpleNet, and a self-designed residual network called MyNet.

Goals

Investigate how well different CNN architectures handle multi-label detection on a small dataset, and demonstrate that transfer learning from ImageNet substantially outperforms training from scratch when labelled data is limited.

Objectives

- Build a data pipeline that produces 224x224 uint8 image tensors with 20-dimensional present/absent label vectors from the VOC 2007 annotations.
- Train AlexNet from scratch and compare it against the same architecture initialised with ImageNet weights and fine-tuned at a lower learning rate.
- Design and train MyNet — a compact residual network with batch normalisation — without any pre-training, and compare it to the scratch-trained baselines.
- Evaluate all models by mAP on the VOC 2007 test split after 20 epochs on an RTX 3080.
- Analyse results to identify the contribution of pre-training, architecture choice, and regularisation (batch norm, residual connections) to final mAP.

Approach

Images are resized to 224x224, normalised with ImageNet channel statistics per batch, and augmented with random horizontal flips during training. The pretrained AlexNet's final classification layer is replaced with a 20-output sigmoid head and fine-tuned end-to-end at a reduced learning rate to preserve low-level ImageNet features. MyNet introduces residual connections and batch normalisation to stabilise training from a random initialisation. All models are trained for 20 epochs and assessed on the held-out test split using per-class AP averaged across all 20 VOC categories.

Outcomes

- ImageNet-pretrained AlexNet (fine-tuned) achieved a test mAP of 0.717 — by far the best result, confirming that transfer learning is decisive when training data is scarce.
- MyNet (self-designed residual net, no pre-training) reached mAP 0.434, outperforming both from-scratch baselines and validating the benefit of batch normalisation and residual connections.
- SimpleNet from scratch achieved mAP 0.308 and AlexNet from scratch 0.196, both hampered by the limited size of the VOC training set.
- The gap between pretrained fine-tuning (0.717) and the best scratch model (0.434) quantifies the value of ImageNet features for a roughly 5,000-image training set.